

Memo 3

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Summary

This memo reports the final status of the summer semester of the MSPH final the project examining dietary choline intake, BMI, and liver enzyme levels in U.S. adults (NHANES August 2021 to August 2023). The project is on track and will continue through the fall.

The question and design are settled, the data are obtained and cleaned, the analytic sample is characterized, and the analysis code is written.

Aim 1 is complete and its results are reported below. What remains is running and interpreting the Aim 2 models and writing up the findings.

Two decisions have changed since the last checkpoint: I am not preregistering on OSF, and I am no longer treating Aim 2 as conditional on a feasibility check.

For full access to the datasets, code, and plots, please visit this github repository: <https://github.com/LouieNeri/MSPH-Final-Project-Choline-BMI>

Final Research Question

Among U.S. adults, does the prevalence of dietary choline intake below the Adequate Intake (AI) differ by BMI category, and is the association between choline intake and liver enzyme levels (ALT, AST, GGT) modified by BMI?

- **Aim 1 (primary):** whether the proportion of adults with choline intake below the AI differs by BMI category, stratified by sex and menopausal status.
- **Aim 2 (secondary):** whether BMI modifies the association between choline intake and liver enzyme levels.

Progress Against Planned Next Steps

1. Preregistration and the transparency approach

Decision: not preregistering on OSF. The prior plan was to preregister before analyzing the data. I decided the formal OSF process was more burdensome than warranted for a project of this scope and timeline.

Instead, the work is kept transparent through version-controlled, shared analysis code. The data pull, the cleaning and exclusion rules, and every model are written out in scripts that anyone can inspect and rerun, in a public repository.

<https://github.com/LouieNeri/MSPH-Final-Project-Choline-BMI/>

2. Obtain, clean, and merge the data; assess the analytic sample

The NHANES August 2021 to August 2023 files were obtained directly through the `nhanesA` package, merged, and cleaned, with the study variables derived and inclusion/exclusion rules applied. The survey design is declared on the full sample and the analytic samples are taken as design subsets, following NCHS guidance, so that the stratum and PSU structure is retained for variance estimation.

Exclusions.

Stage	N
Full merged file	11,933
Adults aged 20+	7,809
Aim 1 analytic sample	4,238
Aim 1 sample used for BMI comparisons	4,160
Aim 2 analytic sample	3,851

The large drop into the Aim 1 sample is mostly participants without two valid dietary recall days, which is expected given that usual-intake estimation requires both days. The further drop to 4,160 removes underweight adults ($n = 78$), whose cell was too small to support a stable weighted estimate; all BMI comparisons below use this sample.

Inclusion criteria: The Aim 1 sample includes non-pregnant, non-lactating adults aged 20 and older with two valid 24-hour dietary recalls and a measured BMI.

Exclusion criteria: Participants with implausible energy intake on either recall day (below 500 or above 8,000 kcal/day for men; below 500 or above 5,500 for women) are excluded.

The Aim 2 sample applies the same criteria and additionally requires a measured ALT value. It excludes participants with active hepatitis, self-reported liver disease, or heavy alcohol use (more than two drinks per day on average for men, more than one for women), because these conditions raise liver enzymes independently of diet and would confound the choline-enzyme association.

Baseline characteristics are shown in Table 2. The sample is 2,324 women and 1,836 men, mean age 48.9 years (SD 17.2). Weighted mean choline intake is 335.0 mg/day overall, 287.4 among women and 384.5 among men. Because the AI is also sex-specific (425 mg/day for women, 550 for men), the sexes end up with similar shortfall prevalence despite the large difference in absolute intake. Two in five adults in the sample are obese (41.9% weighted).

Baseline characteristics of the Aim 1 analytic sample, by sex.

3. Run Aim 1 (primary) and Aim 2

Aim 1 is complete; results are in the next section.

Decision: proceeding with Aim 2 unconditionally. Aim 2 was originally gated behind a feasibility check on the choline and BMI distributions, to be evaluated without looking at the liver enzyme outcomes. I am dropping the gate and fitting the interaction models regardless. The reasoning is that a null or imprecise interaction is itself an informative result for this question. It may be more feasible to also have more detailed conversations around this model, as this would require a bit more technical nuance.

4. Draft the introduction

I did not draft the introduction, as I spent most of my time ensuring that the data was properly cleaned and analyzed for the initial prevalence results.

Table 2: Baseline characteristics of the Aim 1 analytic sample, by sex. Unweighted n with survey-weighted percentages; weighted mean (SD) for continuous variables.

Characteristic	Sex		Overall, N = 4160 ¹	p-value ²
	Female, n = 2324 ¹	Male, n = 1836 ¹		
Age, years	49.6 (17.3)	48.3 (17.0)	48.9 (17.2)	0.12
Race/ethnicity				0.5
Mexican American	143 (6.7%)	112 (7.6%)	255 (7.1%)	
Other Hispanic	226 (9.1%)	170 (9.9%)	396 (9.5%)	
Non-Hispanic White	1,452 (61.4%)	1,160 (62.6%)	2,612 (62.0%)	
Non-Hispanic Black	276 (12.7%)	196 (9.8%)	472 (11.3%)	
Non-Hispanic Asian	95 (5.2%)	78 (5.2%)	173 (5.2%)	
Other Race - Including Multi-Racial	132 (4.9%)	120 (5.0%)	252 (5.0%)	
Education				
Less than 9th grade	83 (2.6%)	73 (3.1%)	156 (2.9%)	
9–11th grade (Includes 12th grade with no diploma)	141 (5.4%)	141 (6.4%)	282 (5.9%)	
High school graduate/GED or equivalent	429 (22.3%)	388 (25.6%)	817 (23.9%)	
Some college or AA degree	757 (30.5%)	509 (29.0%)	1,266 (29.8%)	
College graduate or above	914 (39.1%)	725 (35.9%)	1,639 (37.6%)	
Don't know	0 (0.0%)	0 (0.0%)	0 (0.0%)	
BMI category				0.095
normal	588 (26.0%)	430 (23.1%)	1,018 (24.6%)	
overweight	692 (31.1%)	668 (36.0%)	1,360 (33.5%)	
obese	1,044 (42.8%)	738 (40.9%)	1,782 (41.9%)	
Choline intake, mg/day	287.4 (133.3)	384.5 (184.0)	335.0 (167.3)	<0.001
Below choline AI	1,973 (84.9%)	1,539 (83.2%)	3,512 (84.1%)	0.2

¹ Unweighted n (weighted %) for categorical variables; weighted mean (SD) for continuous variables.

² Wilcoxon rank-sum test for complex survey samples; chi-squared test with Rao & Scott's second-order correction

Table 3: Prevalence of choline intake below the AI, by BMI category.

Prevalence of choline intake below the AI by BMI category	
Unweighted count below AI, survey-weighted % (95% CI)	
BMI category	Overall (N = 4,160)
normal	842, 82.7% (79.8–85.7)
overweight	1143, 84.9% (82.4–87.3)
obese	1527, 84.2% (80.6–87.8)

Aim 1 Results

All estimates are survey-weighted and account for the NHANES stratified multistage design. Cells report the unweighted count below the AI with the weighted percentage and its 95% confidence interval.

Choline intake below the AI is close to universal in this population. Overall, 84.1% of adults (3,512 of 4,160) had usual intake below the sex-specific AI.

Prevalence by BMI category is shown in Table 3 and Figure 1. The three categories are within about two percentage points of one another (normal 82.7%, overweight 84.9%, obese 84.2%) and the confidence intervals overlap substantially.

The sex-stratified results are in Table 4 and Figure 2. The two sexes trend in opposite directions across BMI category. Among women, prevalence rises from 80.1% at normal weight to 87.0% among those with obesity; among men it falls from 85.8% to 81.1%. The confidence intervals overlap at every category.

Among women, prevalence is further stratified by menopausal status in Table 5 and Figure 3. This comparison uses 1,610 of the 2,324 women; the remainder had a hysterectomy or a status that could not be determined. Premenopausal women at normal weight have the lowest prevalence observed anywhere in the analysis (76.6%), and premenopausal prevalence rises steeply with BMI (76.6%, 83.1%, 88.8%). The postmenopausal prevalence are generally similar, (83.2%, 89.0%, 82.2%), with generally wide confidence intervals.

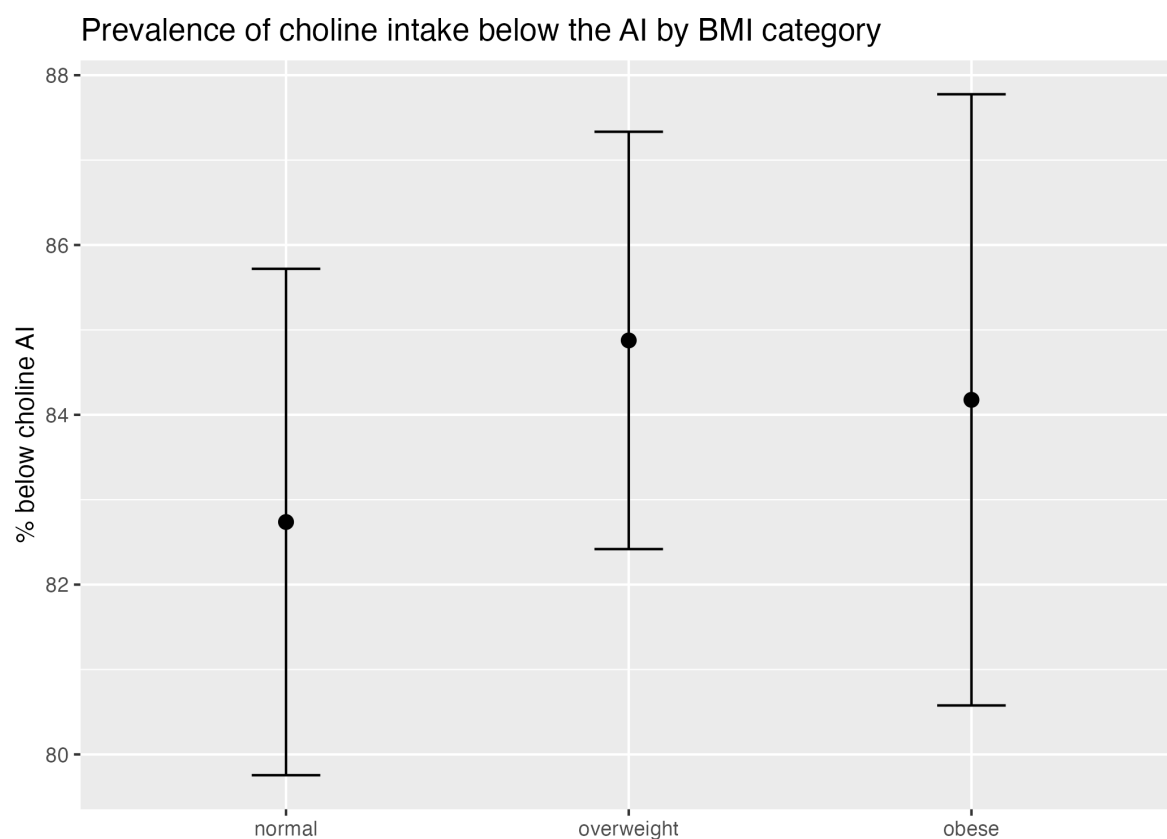


Figure 1: Prevalence of choline intake below the AI, by BMI category. Points are weighted percentages; bars are 95% confidence intervals.

Table 4: Prevalence of choline intake below the AI, by BMI category and sex.

Prevalence of choline intake below the AI		
Unweighted count below AI, survey-weighted % (95% CI)		
BMI category	Female (N = 2,324)	Male (N = 1,836)
normal	483, 80.1% (77.3–83.0)	359, 85.8% (81.0–90.6)
overweight	585, 85.9% (81.9–89.9)	558, 84.0% (81.2–86.8)
obese	905, 87.0% (82.3–91.7)	622, 81.1% (78.3–83.9)

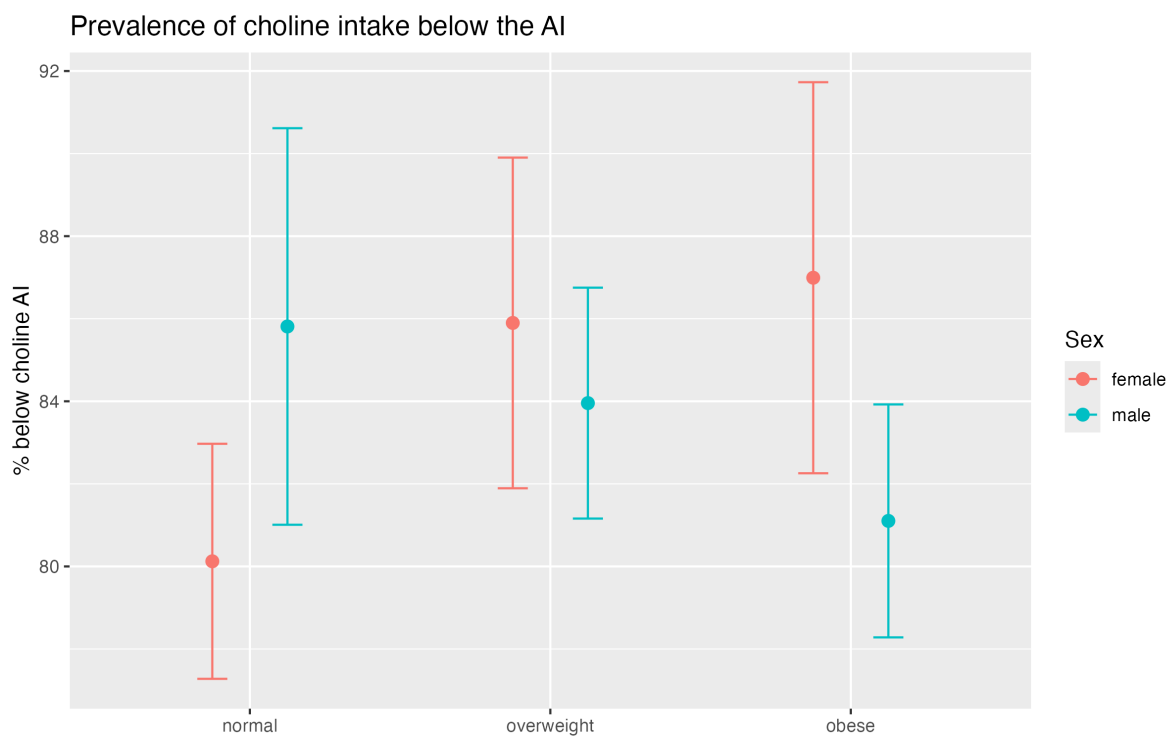


Figure 2: Prevalence of choline intake below the AI, by BMI category and sex.

Table 5: Prevalence of choline intake below the AI among women, by BMI category and menopausal status.

Prevalence of choline intake below the AI among women		
Unweighted count below AI, survey-weighted % (95% CI)		
BMI category	Pre-menopausal (N = 680)	Post-menopausal (N = 930)
normal	167, 76.6% (71.8–81.4)	201, 83.2% (79.6–86.9)
overweight	162, 83.1% (76.2–90.0)	251, 89.0% (84.8–93.2)
obese	230, 88.8% (82.2–95.5)	330, 82.2% (74.4–90.1)

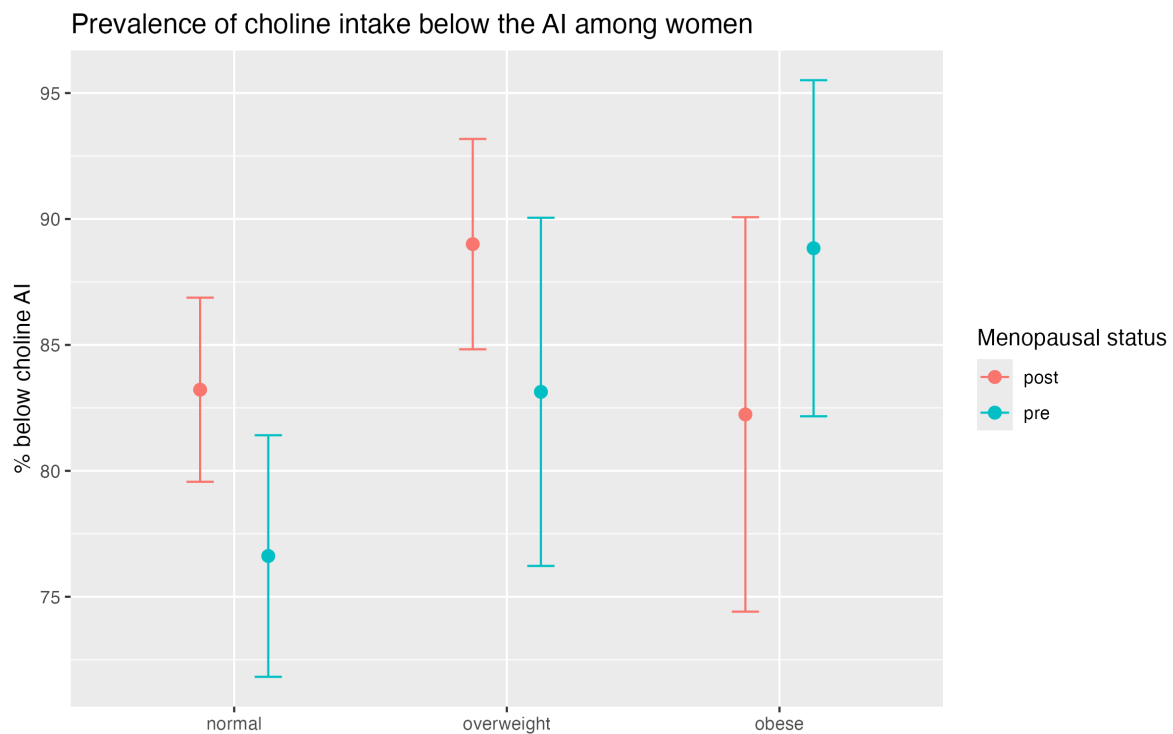


Figure 3: Prevalence of choline intake below the AI among women, by BMI category and menopausal status.

Next Steps: Fall 2026

For the final stretch during August to December 2026. The remaining work is to finish the Aim 2 models, run sensitivity analyses, and write up the project for final submission. With further considerations and additional analyses as conversations go on through the semester.

Plots and tables are currently preliminary, and will be updated for proper formatting.